

WeatherBot

Dashboard Reference Manual

Understanding every element on the station signal cards

This manual explains every field, badge, number, and colour shown on the WeatherBot dashboard cards. Each card represents one city whose daily high temperature is traded on Kalshi prediction markets.

1. Overview — What Is a Signal Card?

The dashboard shows one card per city (station). Each card runs the full signal pipeline and tells you:

- Whether the bot wants to trade, watch, or skip tomorrow's market
- What temperature outcome it is backing and why
- How much it wants to stake, calculated by the Kelly criterion
- Real-time weather conditions and intraday temperature tracking

Cards are refreshed automatically every 6 hours (Tier 3 signal pass) and can be triggered on-demand with the **Run Signal** button.

2. Card Header

2.1 Decision Badge

The coloured badge in the top-right corner of the card is the bot's final verdict for this station. There are five possible values:

TRADE	WATCH	SKIP	HARD SKIP	CONSTRAINED
TRADE	All conditions met — edge clears the threshold, Kelly stake is above the minimum, and capital is available. The bot will place (or has placed) an order.			
WATCH	Edge is positive and the signal is real, but it does not clear today's required threshold. Worth monitoring — conditions may improve later in the day.			
SKIP	No bucket shows positive edge against Kalshi's current prices. The bot sees no mispricing worth trading.			

HARD SKIP	The TAF (terminal forecast) contains dangerous weather — freezing rain, heavy snow, or severe convection. Temperature forecasts become unreliable in these conditions. No trade is ever placed on a HARD SKIP.
CONSTRAINED	The signal is valid and edge is present, but the exposure limit has been reached (another position is already using available capital). The bot will not open a second trade until the first closes.

2.2 Weather Icon Panel

The small panel in the top-left of the card header shows the current live observation (METAR) for the station:

Temperature	Current surface temperature in °F. Colour-coded: blue = cold (<45°F), green = mild, orange = warm (>80°F), red = hot (>95°F).
Sky cover	Current ceiling / cloud cover from the METAR: SKC (clear), FEW, SCT (scattered), BKN (broken), OVC (overcast).
Wind	Wind speed in knots (kt). Direction is the heading the wind is coming FROM, in degrees (e.g. 270° = westerly wind).
Dewpoint	DP = dewpoint temperature. High dewpoint (close to temperature) means humid air — relevant for coastal fog risk stations like KSFO.

If aviationweather.gov is temporarily unavailable, the weather panel is hidden rather than showing stale or invalid data.

2.3 Market Date

Below the header, the label "**High Temperature Market — YYYY-MM-DD**" shows the date of the Kalshi market being analysed. This is always *tomorrow's* date — the bot trades the next-day high temperature market, which settles after the calendar day ends.

3. Forecast Row

3.1 Bias-Adjusted Forecast

The large temperature number labelled "**Bias-adj. forecast**" is the bot's best estimate of tomorrow's high temperature. It is built in two steps:

- **Primary forecast** — fetched live from the NWS api.weather.gov gridded forecast, updated hourly. This is an official NWS human-forecaster product.
- **Bias correction** — for NWS forecasts the correction is zero (NWS forecasters already account for local biases). For ERA5/Open-Meteo fallback sources, a statistical correction is applied from the historical bias table.

The **±X.X°F** figure beside the forecast is the *uncertainty spread* (one standard deviation). It represents the typical error on similar forecast days. For NWS forecasts this is fixed at ±3.5°F — the documented NWS day-ahead mean absolute error. About 68% of similar days land within this range.

3.2 NWS vs MOS Divergence Badge

When GFS-MOS (Model Output Statistics) data is available, a small badge appears showing "**NWS +X°F vs MOS**". This compares the official NWS human forecast against the raw GFS model guidance:

Yellow badge	NWS forecaster is running warmer than the GFS model by $>2^{\circ}\text{F}$. The forecaster may know about local warm advection or sea-breeze patterns the model misses.
Blue badge	NWS forecaster is running cooler than the GFS model by $>2^{\circ}\text{F}$. The forecaster may be accounting for marine influence, cloud cover, or a cold pool the model underestimates.
Grey badge	NWS and GFS agree within $\pm 2^{\circ}\text{F}$ — forecaster and model are aligned.
No badge	GFS-MOS data was not available from IEM today. Signal uses NWS alone.

This badge does NOT change the forecast — it is a confidence indicator only. Large NWS/MOS disagreements widen uncertainty about the outcome.

3.3 Cluster · Season · n=

The small text on the right of the forecast row — e.g. "**Cluster 3 · MAM · n=42**" — describes the synoptic weather regime used to calibrate the forecast:

Cluster N	The 500mb upper-air pattern for today, classified by KMeans into one of several seasonal regimes. Each cluster captures a distinct large-scale weather pattern (e.g. ridge, trough, zonal flow). The bot was trained on historical days with the same cluster.
Season	The meteorological season: DJF (Dec-Jan-Feb), MAM (Mar-Apr-May), JJA (Jun-Jul-Aug), SON (Sep-Oct-Nov). Clusters are trained separately per season because the same upper-air pattern means different things in summer vs winter.
n=	The number of historical observations in the bias table that match today's exact cluster, month, and forecast range. Higher n means the bias correction is more statistically reliable. Below n=10 the bot applies a station/month average fallback instead.

4. Obs Tracking

The **Obs tracking** line shows the current METAR temperature alongside a coloured divergence bubble:

XX°F	The most recent observed temperature at the station (METAR), updated roughly every 20-30 minutes.
+X°F ▲	Current obs are X°F ABOVE the adjusted forecast. Shown in orange/red. A large positive divergence early in the day suggests the high may beat the forecast — good news if you hold a higher bucket.
-X°F ▼	Current obs are X°F BELOW the adjusted forecast. Shown in blue. A large negative divergence mid-afternoon is an early warning that the daily high may fall short — relevant for exit timing.
≈ 0°F	Tracking in line with the forecast (within $\pm 4^{\circ}\text{F}$). No action needed.

Obs tracking disappears when the METAR fetch fails. This is normal and temporary — aviationweather.gov occasionally times out. It returns automatically on the next Tier 1 cycle (every ~5 minutes).

5. Top Bucket Row

The highlighted row labelled **Top bucket** shows the single best market opportunity the bot has identified for this station. It is selected by blending model probability and edge, so it favours the *likely* outcome while still rewarding genuine mispricing by Kalshi.

5.1 Bucket Label

Kalshi's temperature markets are divided into 2°F buckets. The label shows the range the bot is backing:

$\leq 72^{\circ}\text{F}$	The <i>lower tail</i> bucket — settles YES if the daily high is 72°F or below. This is the lowest bucket available in the market.
74–75°F	An <i>interior</i> bucket — settles YES if the daily high falls between 74°F and 75°F (exclusive upper bound).
$\geq 81^{\circ}\text{F}$	The <i>upper tail</i> bucket — settles YES if the daily high is 81°F or above. This is the highest bucket available in the market.

Tail boundaries vary by station and season. They are discovered live from Kalshi's API each cycle — the dashboard always shows the actual live boundaries, not hardcoded constants.

5.2 Model %

The probability the bot's forecast model assigns to this bucket settling YES. Calculated by integrating a normal distribution (mean = adjusted forecast, σ = uncertainty spread) over the bucket's temperature range.

Example: forecast $76^{\circ}\text{F} \pm 3.5^{\circ}\text{F}$ → the 74–75°F bucket covers roughly 22% of the probability distribution.

5.3 Kalshi %

The probability implied by Kalshi's current *ask price* for the Yes contract. If you can buy Yes for 20¢, Kalshi is implying a 20% chance of settlement.

Kalshi % = yes_ask price (e.g. \$0.20 → 20%). This is the price you pay per contract, not the mid-market probability.

5.4 Edge

Edge = Model % – Kalshi %. This is the core signal:

Positive edge	Our model thinks this outcome is MORE likely than Kalshi does. We have a statistical advantage — the market is underpricing the outcome.
Negative edge	Kalshi is pricing this outcome higher than our model. No advantage — do not trade.
✓ checkmark	Edge of +0.12 or above — clears the base threshold required to trade (before any weather penalty is applied).

Edge is expressed as a decimal fraction (e.g. +0.15 = 15 percentage points of advantage). A +0.15 edge means we believe the true probability is 15pp higher than the Kalshi price implies.

6. Kelly Stake Row

The **Kelly stake** line shows how much the bot wants to risk and at what price:

\$X.XX	Total dollar amount to spend on this trade, calculated by the fractional Kelly criterion. Capped at 2% of the current bankroll per trade to prevent overexposure.
N contracts	Number of Yes contracts to buy. Each contract pays \$1.00 if it settles YES and \$0.00 if it settles NO.
@ \$X.XX	The ask price per contract in dollars (e.g. \$0.18 = 18¢). This is what you pay upfront. Your maximum profit per contract is \$1.00 – ask price.

Kelly Criterion Explained

The Kelly criterion is a formula that sizes bets to maximise long-run bankroll growth. Given:

- p = our model probability of winning
- b = net profit per dollar staked $(= (1 - \text{ask}) / \text{ask})$
- $q = 1 - p$

Kelly fraction $= (p \times b - q) / b$

The bot uses *fractional Kelly* — it never bets more than 2% of bankroll regardless of how large the Kelly fraction computes to. This protects against model errors and extreme market moves.

Kelly stake is further scaled down when the synoptic pattern confidence is **MEDIUM** (75% of Kelly) or **LOW** (50% of Kelly) — unusual weather regimes carry higher forecast uncertainty.

A Kelly stake below \$1.00 triggers a WATCH instead of TRADE — the position would be too small to be worth the transaction cost.

7. Expandable Sections

Three collapsible panels can be opened by clicking the buttons at the bottom of each card:

7.1 Full Reasoning

A four-section plain-English explanation of the signal:

Current Conditions	A summary of the live METAR: temperature, dewpoint, wind, and sky cover at the station right now.
Synoptic Pattern	A description of the 500mb upper-air pattern (cluster). Explains what large-scale weather regime is in place and how confidently it matches the historical training set.

Forecast & Bias	Explains the primary forecast source, any NWS/MOS divergence, the historical bias estimate, and the adjusted forecast with its uncertainty range. For NWS forecasts, notes that no bias correction is applied because NWS forecasts are already human-calibrated.
Market Analysis	States what Kalshi is pricing the top bucket at, what the bot's model says, and the resulting edge. Also states the current edge threshold and whether any weather penalty has been applied.
Decision Rationale	One sentence explaining the final TRADE / WATCH / SKIP decision in plain terms — stake size, edge vs threshold, or reason for skipping.

7.2 Bucket Table

A full table of every live Kalshi bucket for this station, showing:

★ (star)	The bucket selected as the top opportunity.
Bucket	The temperature range label (e.g. 74–75°F, ≤ 72°F, ≥ 81°F).
Model %	Our model's probability for this bucket.
Kalshi %	Kalshi's implied probability (ask price).
Edge	Model % – Kalshi %. Green = positive (we have edge). Red = negative (Kalshi overpricing this outcome).
Bar	A visual bar proportional to the magnitude of edge. Green bar = positive edge; red bar = negative edge.

Buckets with model probability below 75% of the peak bucket are still shown in the table for reference, but are not eligible for selection as the top trading bucket.

7.3 Weather Summary

A detailed breakdown of the current meteorological data:

METAR	The full current observation: temperature, dewpoint, wind direction and speed, sky cover, and the raw METAR string for reference.
TAF Highlights	Key points from the Terminal Aerodrome Forecast for the next 24 hours: ceiling, weather condition category, and whether an Amendment (AMD) has been issued. AMD means the forecast was revised after the original issuance — useful context when conditions are changing rapidly.

8. Weather Condition Categories & Thresholds

The TAF is parsed into one of six condition categories. Each category adjusts the edge threshold required to place a trade — bad weather makes temperature forecasts less reliable, so the bot demands higher edge:

clear	Clear skies, no significant weather. Base threshold applies (typically 12%).
scattered	Scattered clouds (3–4 oktas). Mild penalty — threshold increases slightly.

broken	Mostly cloudy / broken ceiling. Moderate penalty — forecasts less certain due to reduced solar heating.
marine_fog	Marine layer or coastal fog risk (relevant for KLAX, KSFO, KSEA). Significant penalty — fog burns off unpredictably and can cap the daily high.
convective	Thunderstorm activity in the TAF. High penalty — convective initiation timing is very difficult to forecast.
precip	Active precipitation (rain, drizzle). High penalty — evaporative cooling suppresses the daily high in ways models often underestimate.
hard_skip	Dangerous conditions: freezing rain, heavy snow, or ice. The bot will not trade regardless of edge — temperature forecasts under these conditions are considered unreliable. Decision = HARD SKIP.

9. Threshold Checklist

Inside the Full Reasoning panel, a checklist shows whether each guardrail passed or failed:

Weather condition	Passes if the TAF does not trigger a HARD SKIP. A failed check here means no trade is placed regardless of other factors.
Edge vs threshold	Passes if the top bucket's edge meets or exceeds the weather-adjusted threshold. The threshold is the base rate (12%) plus any weather penalty.
Bias uncertainty gate	Passes if the forecast uncertainty (bias_std) is below a level that would make the probability distribution too flat to trade confidently. A very wide spread means almost all buckets have similar probability, making edge hard to find reliably.
Minimum stake	Passes if the Kelly-calculated stake is at least \$1.00. Below this the trade is not worth opening.
Historical sample size	Passes if the bias table has at least 10 observations for today's regime. Below 10 the bias correction is less statistically reliable.
Pattern confidence	HIGH = today matches the cluster centroid closely. MEDIUM = reasonable match. LOW = unusual synoptic setup. Low confidence does not prevent trading but reduces the Kelly stake to 50%.

10. Operating Modes

The bot runs in one of three modes, shown in the top-right of the dashboard header:

DEMO	Paper trading. All logic runs normally but orders are simulated — no real money is placed on Kalshi. P&L; is tracked against a virtual starting bankroll. This is the default mode.
LIVE (dry run)	Connected to the live Kalshi API and reading real market data, but still not placing real orders. Useful for validating API connectivity and order logic before going fully live.

LIVE TRADING	Fully live. Real orders are placed on Kalshi using real funds. All risk limits (2% max stake, 50% max exposure, stop-loss) apply.
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Mode is set by environment variables in the .env file: USE_DEMO=true/false and DRY_RUN=true/false.

11. Signal Refresh Schedule

Tier 1 (~5 min)	METAR fetch, intraday temperature tracking, position exit checks (stop-loss, profit target, overshoot/undershoot guards).
Tier 2 (~10 min)	TAF monitor — checks for amended forecasts that might change the weather penalty category on open positions.
Tier 3 (6 hrs)	Full signal recompute for all 20 stations: new NWS forecast, pattern classification, bias lookup, Kalshi price fetch, edge calculation, Kelly sizing. This is what populates the cards.
Settlement (09:00 UTC)	Checks Kalshi settlement results for yesterday's markets. Closes any positions that settled and records P&L.;
Obs update (10:00 UTC)	Appends yesterday's observed high temperatures from NOAA CDO to the historical database — keeps the bias table current.
Pipeline (1st of month)	Full rebuild of the 500mb pattern clusters and bias table using the most recent month of observations.

Tier 3 can also be triggered on-demand via the Run Signal button on the dashboard.

12. Key Terms Glossary

Kalshi bucket	A 2°F temperature range that defines one prediction market. Buying Yes on a bucket pays \$1.00 if the official daily high falls within that range, \$0.00 otherwise.
Edge	The gap between our model's probability and Kalshi's implied probability. Positive edge means we think the outcome is more likely than the market does.
Bias correction	A statistical adjustment to a raw model forecast based on how much that model historically over- or under-predicts in similar conditions.
Kelly criterion	A mathematical formula for optimal bet sizing. Maximises long-run bankroll growth by betting proportionally to edge.
500mb / z500	The geopotential height of the 500 millibar pressure level in the atmosphere (~18,000 ft). Used to classify the large-scale weather pattern (synoptic regime) driving surface temperatures.
METAR	Aviation weather observation issued every 20-60 minutes. Contains current temperature, dewpoint, wind, visibility, and sky cover.
TAF	Terminal Aerodrome Forecast — a 24-30 hour aviation weather forecast issued by the NWS. Used to assess weather penalty category.
NWS AFM	Area Forecast Matrix — a gridded product issued by NWS human forecasters containing their official max/min temperature guidance.

GFS-MOS	GFS Model Output Statistics — a statistically post-processed version of the GFS global model. Used as a cross-check against the NWS forecast.
Settlement station	The specific NWS weather station whose official daily maximum temperature Kalshi uses to settle the market. Not always the same as the airport ICAO code (e.g. KORD markets settle on KMDW — Chicago Midway, not O'Hare).